

Matematika Sains

DATA

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PARAMETER PERSONALISASI

- $N \text{ (NIM mod 100)}$

$$N = 19 \text{ mod } 100 = \underline{\underline{19}} \rightarrow 19 : 100 = 0 \text{ sisa } (19)$$

- $a \text{ (Floor (N/10))}$

$$a = \lfloor 19 / 10 \rfloor = 1,9 \approx \underline{\underline{1}} \text{ (pembulatan ke bawah)}$$

- $b \text{ (N mod 10)}$

$$b = 19 \text{ mod } 10 = \underline{\underline{9}} \rightarrow 19 : 10 = 1 \text{ sisa } (9)$$

- $K \rightarrow (a+b+1)$

$$K = 1 + 9 + 1 = \underline{\underline{10}}$$

- $\theta \rightarrow (a \cdot 30) + (b \cdot 5)$

$$\theta = (1 \cdot 30) + (9 \cdot 5) = 30 + 45 = \underline{\underline{75^\circ}}$$

- $\alpha \rightarrow 0,001 \cdot (b+1)$

$$\alpha = 0,001 \cdot (9+1) = 0,001 \times 10 = \underline{\underline{0,01}}$$

5) a) Limit aljabar

- L1 (pembuktian)

$$\lim_{u \rightarrow 11} \frac{u^2 - 11^2}{u - 11} = \lim_{u \rightarrow 11} \frac{(u+11)(u-11)}{u-11} = \lim_{u \rightarrow 11} (u+11) = 11+11 = 22 //$$

- L2 (kali sekawan)

$$\lim_{u \rightarrow 0} \frac{\sqrt{u+(11+1)^2} - (11+1)}{u} = \lim_{u \rightarrow 0} \frac{\sqrt{u+12^2} - 12}{u} = \lim_{u \rightarrow 0} \frac{\sqrt{u+144} - 12}{u} \cdot \frac{\sqrt{u+144} + 12}{\sqrt{u+144} + 12}$$

$$= \lim_{u \rightarrow 0} \frac{u+144 - 144}{u(\sqrt{u+144} + 12)} = \lim_{u \rightarrow 0} \frac{u}{u(\sqrt{u+144} + 12)} = \frac{1}{\sqrt{0+144} + 12} = \frac{1}{12+12} = \frac{1}{24} //$$

- L3

$$\begin{aligned} \lim_{u \rightarrow \infty} \frac{(1+2)u^3 + 9u + 7}{u^3 - u + 11} &= \lim_{u \rightarrow \infty} \frac{3u^3 + 9u + 7}{u^3 - u + 11} = \lim_{u \rightarrow \infty} \frac{\frac{3u^3}{u^3} + \frac{9u}{u^3} + \frac{7}{u^3}}{\frac{u^3}{u^3} - \frac{u}{u^3} + \frac{11}{u^3}} = \lim_{u \rightarrow \infty} \frac{3 + \frac{9}{u^2} + \frac{7}{u^3}}{1 - \frac{1}{u^2} + \frac{11}{u^3}} \\ &= \frac{3 + \frac{9}{\infty^2} + \frac{7}{\infty^3}}{1 - \frac{1}{\infty^2} + \frac{11}{\infty^3}} = \frac{3+0+0}{1-0+0} = 3 // \end{aligned}$$

c) $\lim_{u \rightarrow 0} \frac{\sin((1+1)u) \cdot \tan((9+2)u)}{u^2} = \lim_{u \rightarrow 0} \frac{\sin(2u) \cdot \tan(11u)}{u \cdot u}$ y bagi menjadi 2 bagian

$$\left(\lim_{u \rightarrow 0} \frac{\sin(2u)}{u} \right) \cdot \left(\lim_{u \rightarrow 0} \frac{\tan(11u)}{u} \right) = 2 \cdot 11 = 22 //$$

∴ Pendekatan ini absah berdasarkan teorema Limit Trigonometri, dimana $\lim_{u \rightarrow 0} \frac{\sin(au)}{u} = a$

dan $\lim_{u \rightarrow 0} \frac{\tan(bu)}{u} = b$. Serta menggunakan hukum perkalian limit.

$$d) R(n) = \frac{100n + 11}{n + (1+9+1)} = \frac{100n + 11}{n + 11}$$

$$\lim_{n \rightarrow \infty} \frac{100n + 11}{n + 11} = \lim_{n \rightarrow \infty} \frac{\frac{100n}{n} + \frac{11}{n}}{\frac{n}{n} + \frac{11}{n}} = \lim_{n \rightarrow \infty} \frac{100 + \frac{11}{n}}{1 + \frac{11}{n}} = \frac{100 + \frac{11}{\infty}}{1 + \frac{11}{\infty}} = 100 //$$

$\therefore$ Dalam jangka panjang, pendapatan bulanan dari pelanggan reguler akan stabil di angka 100 unit mata uang //

$$e) \lim_{n \rightarrow \infty} \sqrt{n^2 + 11n} - \sqrt{n^2 + 9n}$$

$$\lim_{n \rightarrow \infty} \frac{(\sqrt{n^2 + 11n} - \sqrt{n^2 + 9n})(\sqrt{n^2 + 11n} + \sqrt{n^2 + 9n})}{\sqrt{n^2 + 11n} + \sqrt{n^2 + 9n}} = \lim_{n \rightarrow \infty} \frac{(n^2 + 11n) - (n^2 + 9n)}{\sqrt{n^2 + 11n} + \sqrt{n^2 + 9n}} = \lim_{n \rightarrow \infty} \frac{2n}{\sqrt{n^2 + 11n} + \sqrt{n^2 + 9n}}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{2n}{n}}{\frac{\sqrt{n^2 + 11n}}{\sqrt{n^2}} + \frac{\sqrt{n^2 + 9n}}{\sqrt{n^2}}} = \lim_{n \rightarrow \infty} \frac{2}{\sqrt{1 + \frac{11}{n}} + \sqrt{1 + \frac{9}{n}}} = \frac{2}{\sqrt{1 + \frac{11}{\infty}} + \sqrt{1 + \frac{9}{\infty}}} = \frac{2}{\sqrt{1+0} + \sqrt{1+0}} = \frac{2}{1+1} = \frac{2}{2} = 1 //$$

$\therefore$ Meskipun 2 model ELV terlihat berbeda, dalam jangka panjang keduanya akan stabil pada angka 1, menunjukkan bahwa pertumbuhan bisnis lebih berpengaruh daripada //

f) Syarat kontinu di $u = K \rightarrow u = 11$

① Nilai fungsi terdefinisi

Nilai $C(K)$ dalam soal ini sebagai P .

② Nilai limit ada ($u < K$)

• Limit kiri

$$\lim_{u \rightarrow K^-} \frac{u^2 - K^2}{u - K} = \lim_{u \rightarrow K^-} \frac{(u+K)(u-K)}{\cancel{u-K}} = \lim_{u \rightarrow K^-} u+K = K+K = 2K$$

• Limit kanan ($u > K$)

$$\lim_{u \rightarrow K^+} (a \cdot u + q) = aK + q$$

• Nilai P

Karen nilai $u = K$ adalah $C(K) = P$, maka harus sama dengan limit kiri

$$P = \lim_{u \rightarrow K^-} C(K)$$

$$P = 2K //$$

• Nilai q

Agar kontinu, limit kanan harus sama dengan nilai fungsi P

$$aK + q = P$$

$$aK + q = 2K$$

$$q = 2K - aK$$

$$q = K(2-a) //$$